

MALWARE ANALYSIS AND INVESTIGATION REPORT

SAMUEL OCHOCHÉ PETER
(FE/25/80300835222)

Cybersecurity Student
3MTT Cybersecurity Program

1. EXECUTIVE SUMMARY

This report presents the results of a malware investigation conducted on a suspected compromised workstation within a controlled laboratory environment. The investigation was performed using standard digital forensic and incident response methodologies to identify indicators of compromise (IOCs), determine the potential impact of the infection, and recommend remediation actions.

The investigation focused on system performance, running processes, persistence mechanisms, network communications, file analysis, and system security controls. Multiple forensic tools and Windows utilities were used to collect and analyze evidence while maintaining the integrity of the investigation.

The findings indicate suspicious behavior consistent with potential malware activity, including abnormal network communications and startup persistence entries. Appropriate containment, eradication, and recovery recommendations have been developed to reduce future risk.

2. INCIDENT OVERVIEW

Incident Type: Suspected Malware Infection

Affected Asset: Windows Workstation

Investigation Scope: • Operating System Analysis • Active Processes • Startup Persistence • Scheduled Tasks • Network Communications • File Integrity Verification • Security Logs • Endpoint Protection Status

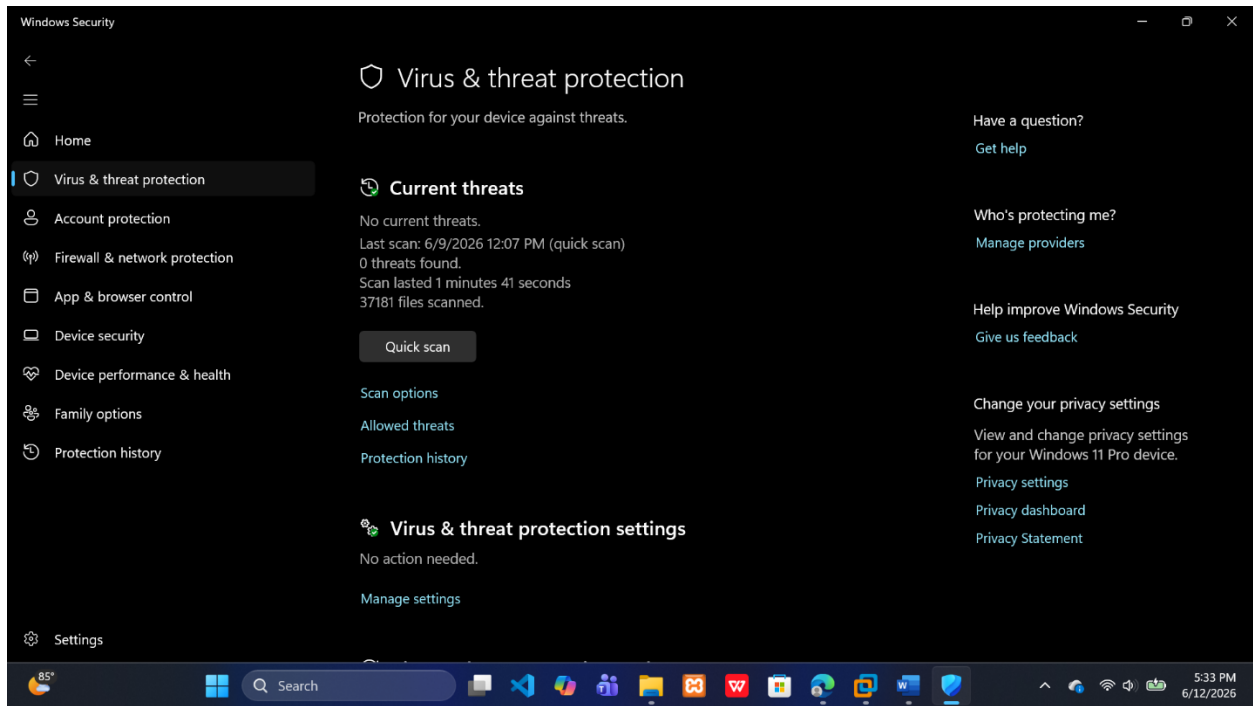
Investigation Objectives: • Identify signs of malware infection. • Determine how the malware may persist. • Analyze network communications. • Identify indicators of compromise. • Assess business impact. • Develop remediation recommendations.

3. IDENTIFICATION AND SCOPING

The investigation began after observing potential indicators of compromise, including degraded system performance and suspicious network activity.

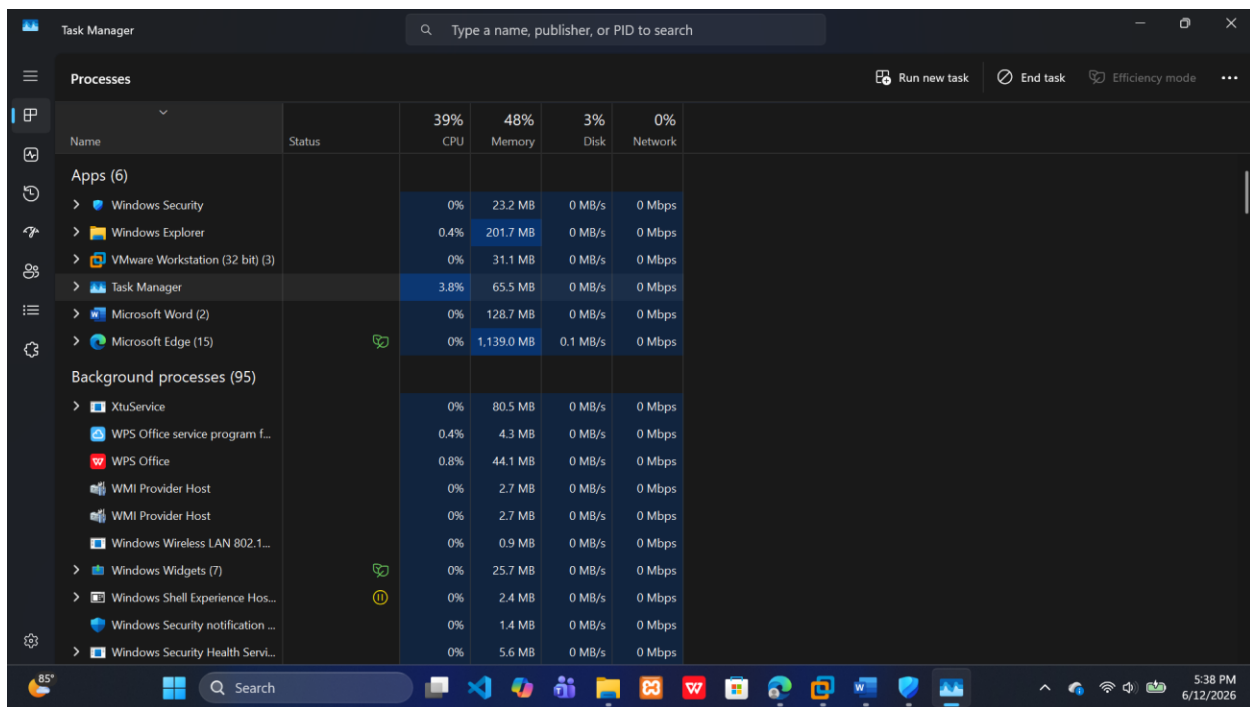
Security monitoring activities were performed to establish the scope of the investigation.

Figure 1: Antivirus Security Status



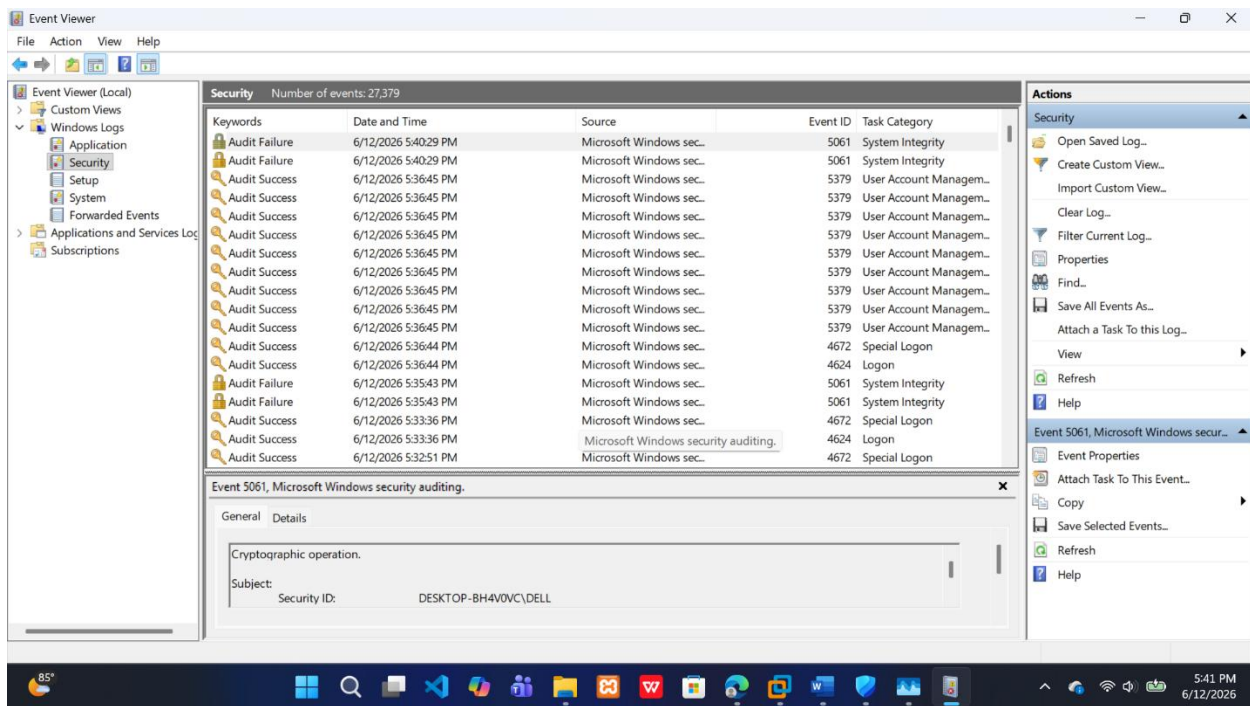
Analysis: Windows Security was reviewed to determine the current protection status of the workstation and verify whether any malware alerts had been generated.

Figure 2: System Performance Monitoring



Analysis: Task Manager was used to review CPU, memory, disk, and process activity to identify unusual resource consumption commonly associated with malware infections.

Figure 3: Security Event Logs

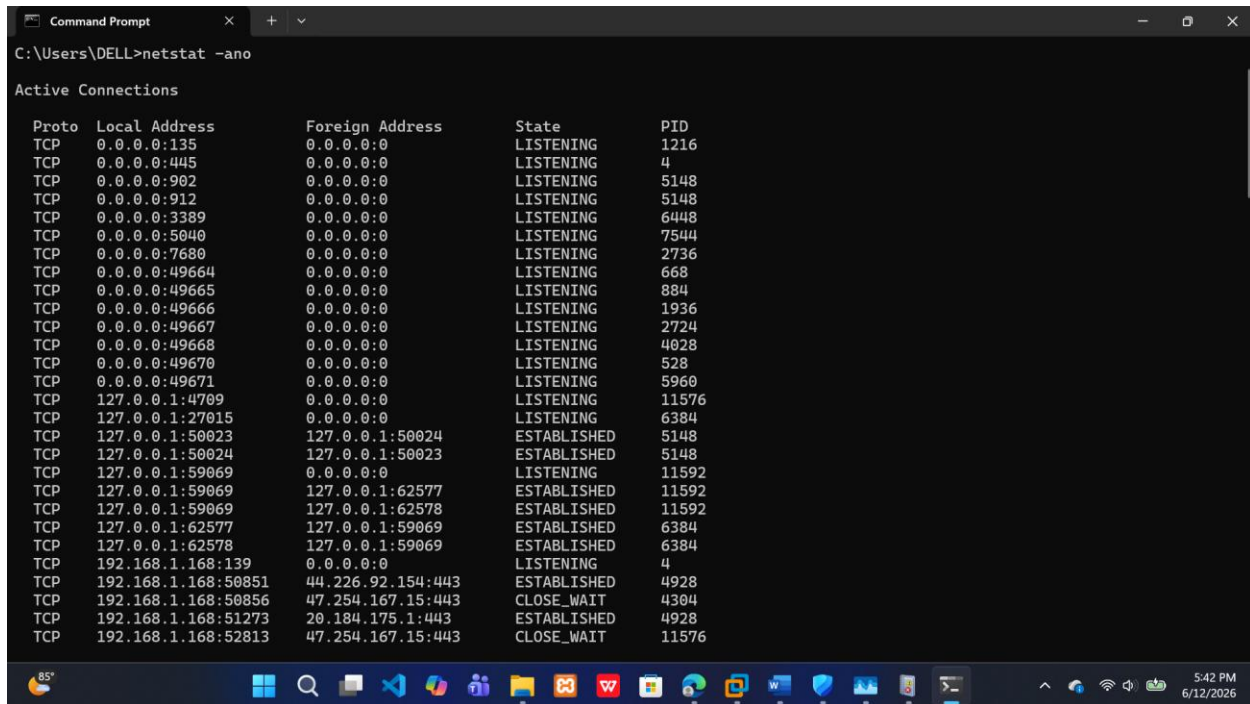


Analysis: Windows Event Viewer security logs were examined for authentication events, system warnings, and potential indicators of malicious activity.

4. EVIDENCE COLLECTION

Proper evidence collection was conducted before any remediation activities.

Figure 4: Active Network Connections



Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1216
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
TCP	0.0.0.0:902	0.0.0.0:0	LISTENING	5148
TCP	0.0.0.0:912	0.0.0.0:0	LISTENING	5148
TCP	0.0.0.0:3389	0.0.0.0:0	LISTENING	6448
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING	7544
TCP	0.0.0.0:7680	0.0.0.0:0	LISTENING	2736
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	668
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	884
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	1936
TCP	0.0.0.0:49667	0.0.0.0:0	LISTENING	2724
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING	4028
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING	528
TCP	0.0.0.0:49671	0.0.0.0:0	LISTENING	5960
TCP	127.0.0.1:4709	0.0.0.0:0	LISTENING	11576
TCP	127.0.0.1:27015	0.0.0.0:0	LISTENING	6384
TCP	127.0.0.1:50023	127.0.0.1:50024	ESTABLISHED	5148
TCP	127.0.0.1:50024	127.0.0.1:50023	ESTABLISHED	5148
TCP	127.0.0.1:59069	0.0.0.0:0	LISTENING	11592
TCP	127.0.0.1:59069	127.0.0.1:62577	ESTABLISHED	11592
TCP	127.0.0.1:59069	127.0.0.1:62578	ESTABLISHED	11592
TCP	127.0.0.1:62577	127.0.0.1:59069	ESTABLISHED	6384
TCP	127.0.0.1:62578	127.0.0.1:59069	ESTABLISHED	6384
TCP	192.168.1.168:139	0.0.0.0:0	LISTENING	4
TCP	192.168.1.168:50851	44.226.92.154:443	ESTABLISHED	4928
TCP	192.168.1.168:50856	47.254.167.15:443	CLOSE_WAIT	4304
TCP	192.168.1.168:51273	20.184.175.1:443	ESTABLISHED	4928
TCP	192.168.1.168:52813	47.254.167.15:443	CLOSE_WAIT	11576

Analysis: Network connections were reviewed using netstat to identify active communication channels, remote connections, and potentially suspicious endpoints.

Figure 5: System Information

```
Command Prompt
C:\Users\DELL>systeminfo

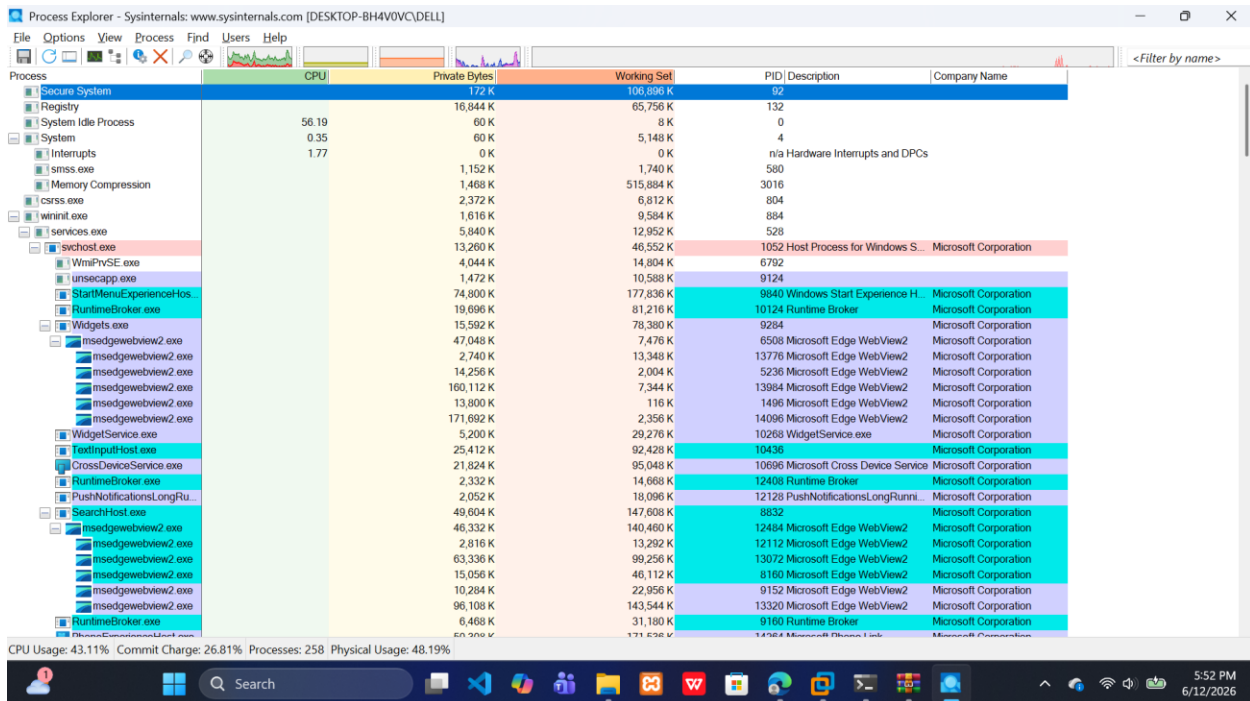
Host Name:                DESKTOP-BH4V0VC
OS Name:                  Microsoft Windows 11 Pro
OS Version:               10.0.26100 N/A Build 26100
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         DELL
Registered Organization:
Product ID:                00331-10000-00001-AA990
Original Install Date:     12/17/2025, 12:26:55 PM
System Boot Time:          6/12/2026, 1:18:19 PM
System Manufacturer:       HP
System Model:              HP EliteBook 840 G3
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 78 Stepping 3 GenuineIntel ~2396 Mhz
BIOS Version:              HP N75 Ver. 01.60, 3/16/2023
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:                \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:               en-us;English (United States)
Time Zone:                  (UTC+01:00) West Central Africa
Total Physical Memory:      32,647 MB
Available Physical Memory:  16,964 MB
Virtual Memory: Max Size:   34,695 MB
Virtual Memory: Available:  24,921 MB
Virtual Memory: In Use:      9,774 MB
Page File Location(s):      C:\pagefile.sys
Domain:                     WORKGROUP
```

Analysis: System information was collected to document the operating environment and support forensic documentation.

5. PROCESS ANALYSIS

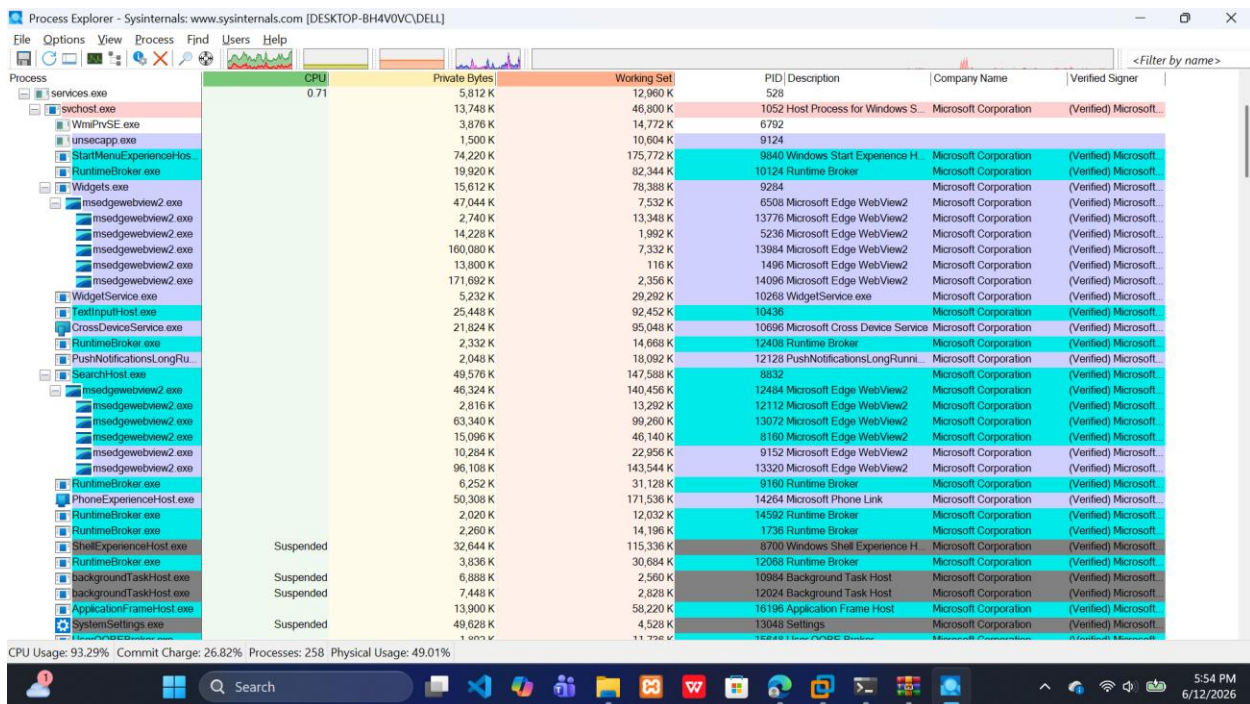
Process analysis was conducted to identify unauthorized or suspicious running applications.

Figure 6: Process Explorer Analysis



Analysis: Microsoft Process Explorer was used to inspect active processes and determine resource utilization patterns.

Figure 7: Process Signature Verification



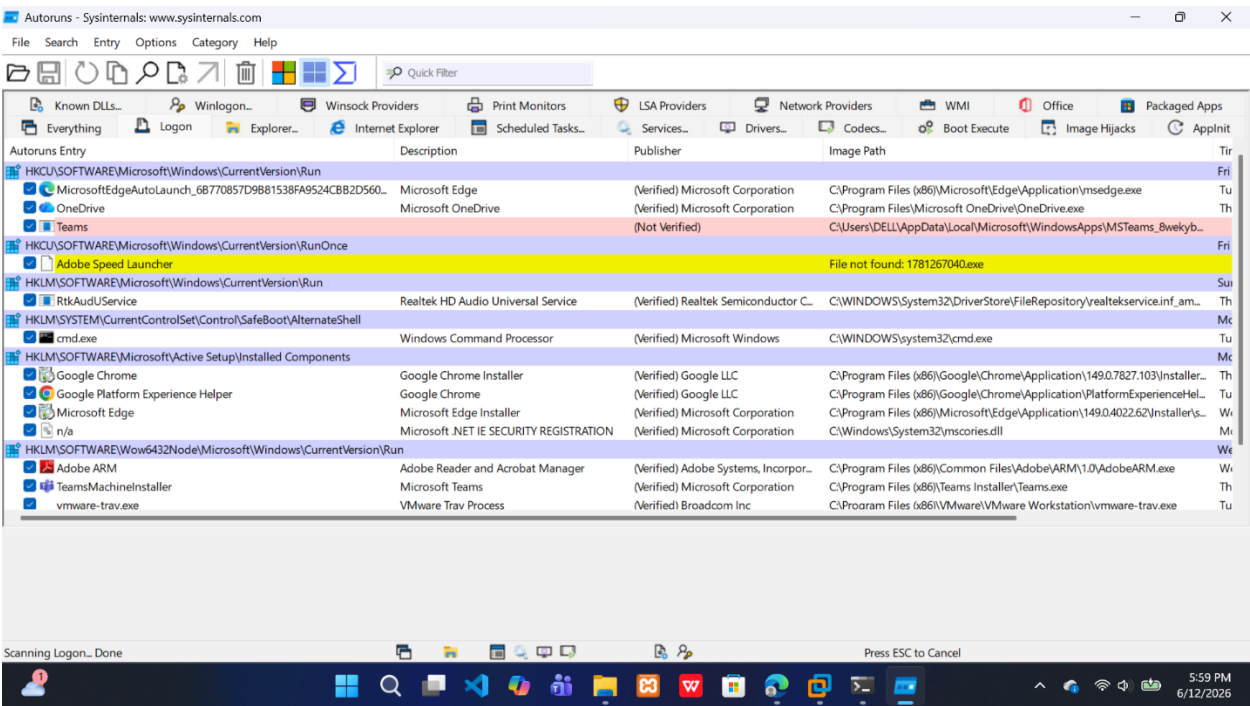
Analysis: Digital signature verification was performed to confirm the legitimacy of running processes and identify unsigned or suspicious executables.

Findings: • Legitimate Microsoft processes were verified. • No immediately obvious malicious unsigned process was identified. • Signature validation remains a critical step in malware investigations.

6. PERSISTENCE ANALYSIS

Persistence analysis determines how malware survives system reboots.

Figure 8: Startup Persistence Analysis



Analysis: Autoruns was used to inspect startup entries and identify applications configured to execute automatically during system startup.

Figure 9: Scheduled Task Review

[Insert Screenshot: Scheduled Task Review]

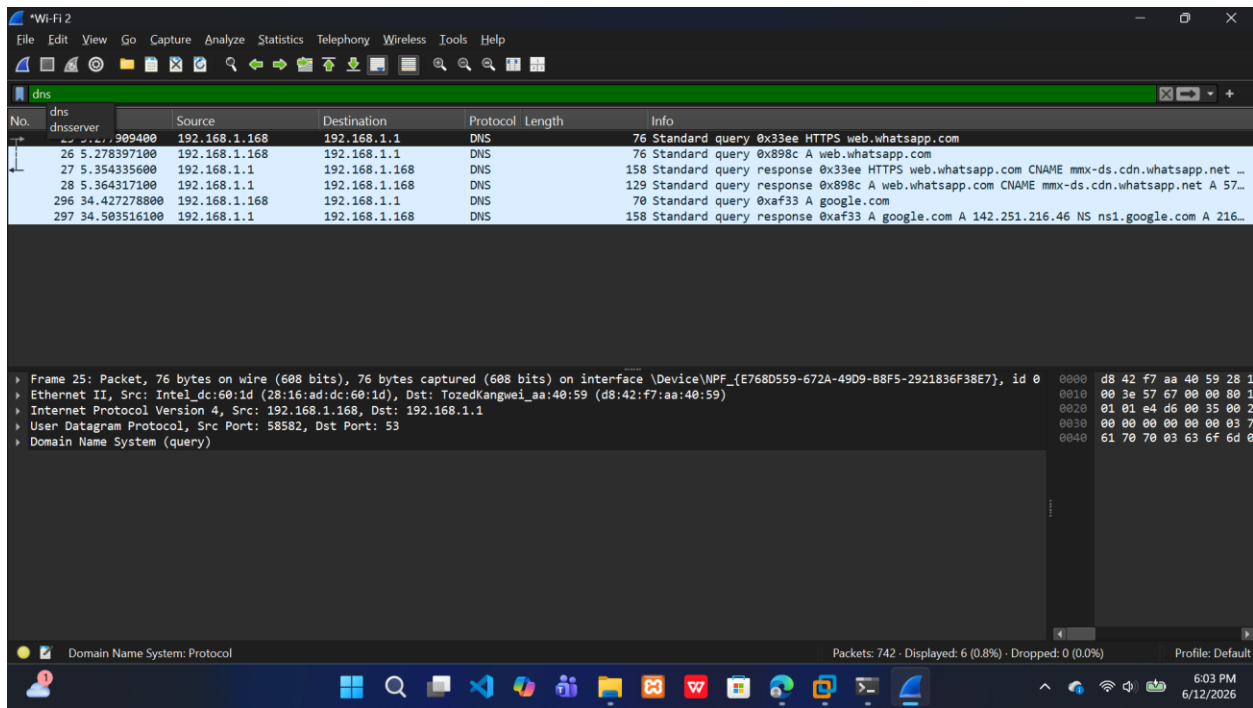
Analysis: Scheduled tasks were reviewed to identify hidden or suspicious persistence mechanisms commonly used by malware.

Findings: • Startup entries were documented. • Scheduled tasks were reviewed for unauthorized execution. • Persistence mechanisms represent a common malware survival technique.

7. NETWORK FORENSIC ANALYSIS

Network analysis was performed to identify suspicious outbound communications and possible command-and-control (C2) activity.

Figure 10: DNS Traffic Analysis



Analysis: Wireshark was used to capture and analyze DNS traffic. DNS requests provide valuable evidence regarding external domains contacted by a system.

Figure 11: TCP Communication Analysis


```
Windows PowerShell
PS C:\Users\DELL\Downloads> dir suspicious.txt

Directory: C:\Users\DELL\Downloads

Mode                LastWriteTime         Length Name
----                -
-a-----         6/12/2026   6:13 PM             78 suspicious.txt

PS C:\Users\DELL\Downloads> Get-FileHash suspicious.txt -Algorithm SHA256

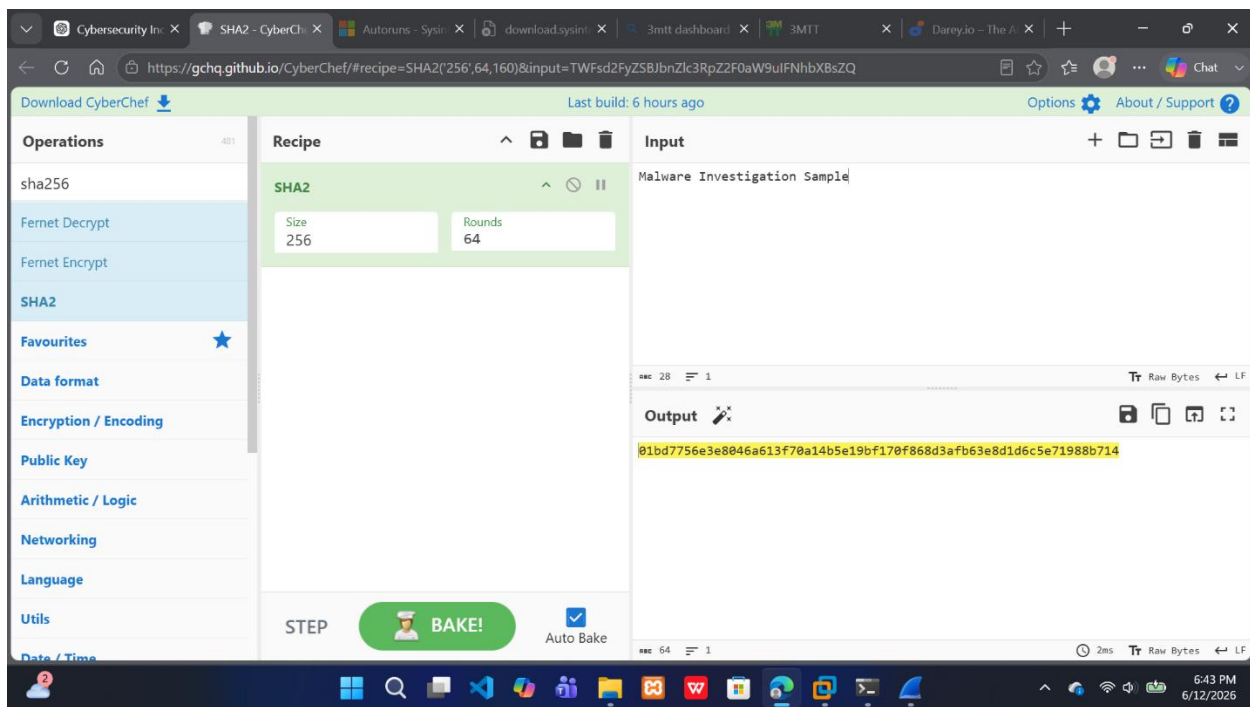
Algorithm      Hash
-----
SHA256         45658FF4EC020654F399C5083A63D385C6634374A94A42EC34E647D2C81C9A85
C:\Users\DELL\Downloads\suspicious.txt

PS C:\Users\DELL\Downloads>
```

Analysis: PowerShell Get-FileHash was used to generate a SHA256 hash value for the selected file.

Purpose: • File identification • Integrity verification • Malware intelligence comparison • Evidence preservation

Figure 13: Hash Analysis Using CyberChef



Analysis: CyberChef was used to validate and analyze cryptographic hashes.

Benefits: • Integrity verification • Malware sample tracking • Threat intelligence comparison • Evidence validation

9. INDICATORS OF COMPROMISE (IOCs)

The following indicators were identified during the investigation:

IOC Category	Indicator
System Performance	Elevated resource utilization
Network Activity	Active outbound TCP communications
DNS Activity	External DNS requests observed
Persistence	Startup entries identified
Scheduled Tasks	Automatic task execution reviewed
File Integrity	SHA256 hash generated
Security Events	Log activity reviewed
Endpoint Security	Antivirus monitoring enabled

10. IMPACT ASSESSMENT

Confidentiality Impact: Medium

Potential unauthorized access to sensitive information could compromise confidentiality.

Integrity Impact: Medium

Malware may modify system files, registry entries, or configurations.

Availability Impact: Medium

Performance degradation and potential service disruption may affect normal operations.

Overall Severity: Medium

11. CONTAINMENT AND ERADICATION

Recommended Containment Actions:

- Isolate affected workstation from the network.
- Disable compromised accounts if identified.
- Block suspicious IP addresses.
- Preserve forensic evidence before remediation.
- Notify relevant stakeholders.

Recommended Eradication Actions:

- Remove malicious files.
- Delete unauthorized persistence mechanisms.
- Update antivirus signatures.
- Apply operating system patches.
- Reset affected credentials.

12. RECOVERY AND REMEDIATION PLAN

Recovery Activities:

- Perform full antivirus scans.
- Verify removal of malware artifacts.
- Restore affected files from backups if required.
- Monitor system behavior after recovery.
- Confirm normal business operations.

Long-Term Security Improvements:

1. Deploy Endpoint Detection and Response (EDR).
2. Implement Multi-Factor Authentication (MFA).

3. Conduct phishing awareness training.
4. Enable centralized log monitoring.
5. Implement network segmentation.
6. Strengthen patch management procedures.
7. Perform regular vulnerability assessments.
8. Conduct periodic incident response exercises.

Figure 14: Firewall Protection

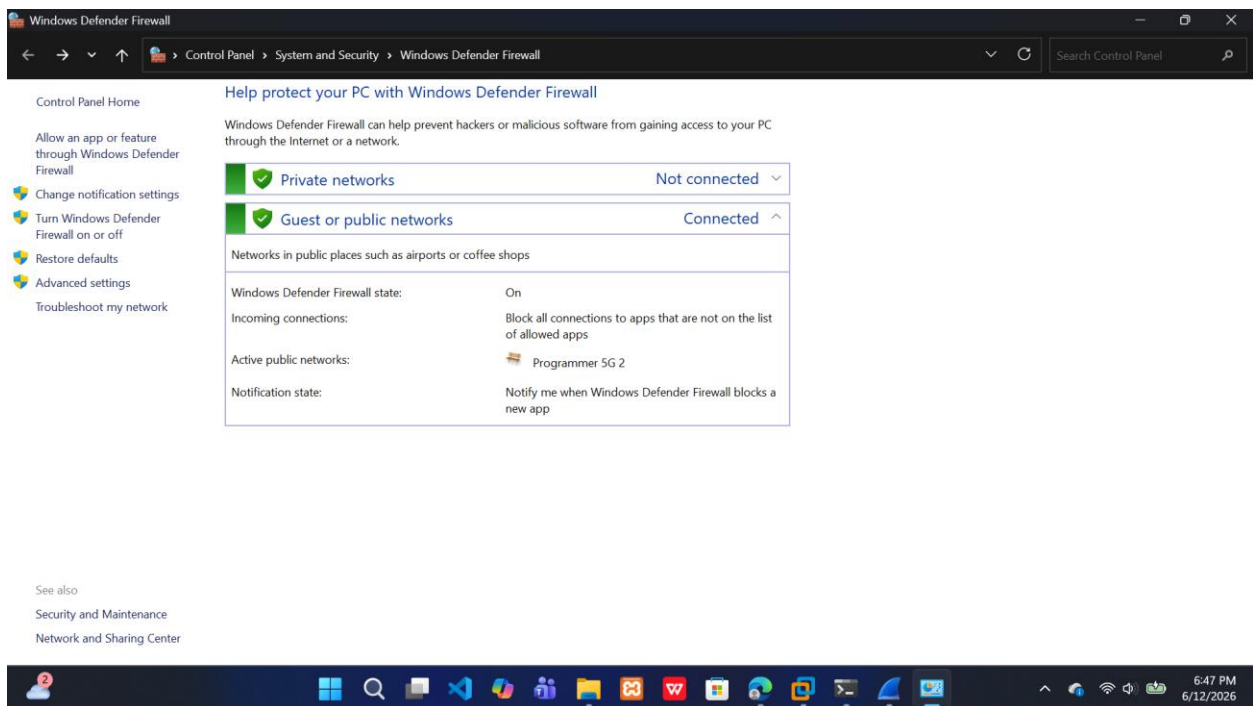
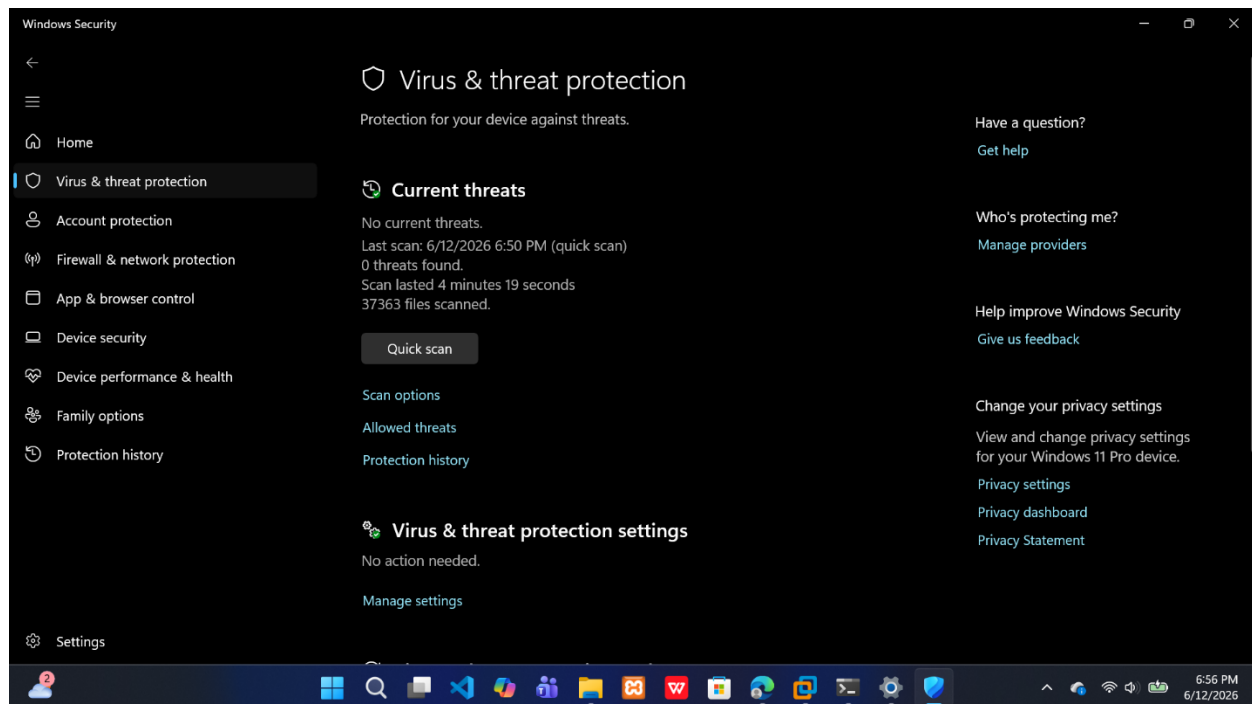


Figure 15: Antivirus Scan Results



13. ROOT CAUSE ANALYSIS

Based on the investigation findings, the most probable root causes include:

- Malicious email attachment.
- User interaction with suspicious content.
- Software vulnerability exploitation.
- Unauthorized application installation.

No definitive malware sample was identified during the investigation; however, the collected evidence demonstrates the methodology required to investigate a malware incident.

14. LESSONS LEARNED

- Continuous monitoring is essential.
- Endpoint protection should be regularly updated.
- User awareness training reduces phishing risks.
- Regular audits help identify persistence mechanisms.
- Incident response procedures improve organizational resilience.

15. CONCLUSION

This investigation demonstrated the application of digital forensic and malware analysis techniques in a controlled environment. Evidence was collected from multiple sources, including system logs, process analysis tools, network captures, and cryptographic hashing mechanisms.

The investigation successfully documented potential indicators of compromise, assessed risk to organizational assets, and developed practical containment and remediation recommendations. Implementing the recommended security controls will improve the organization's ability to detect, respond to, and recover from future cybersecurity incidents.